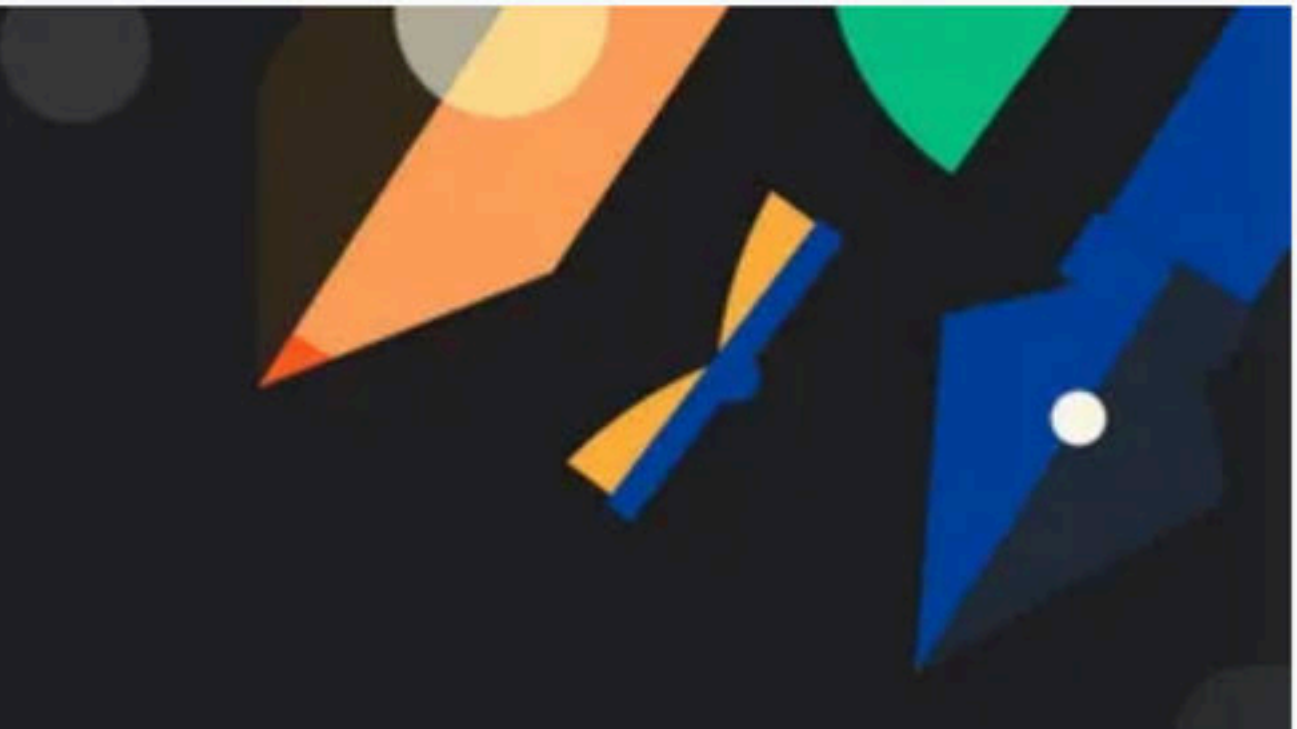
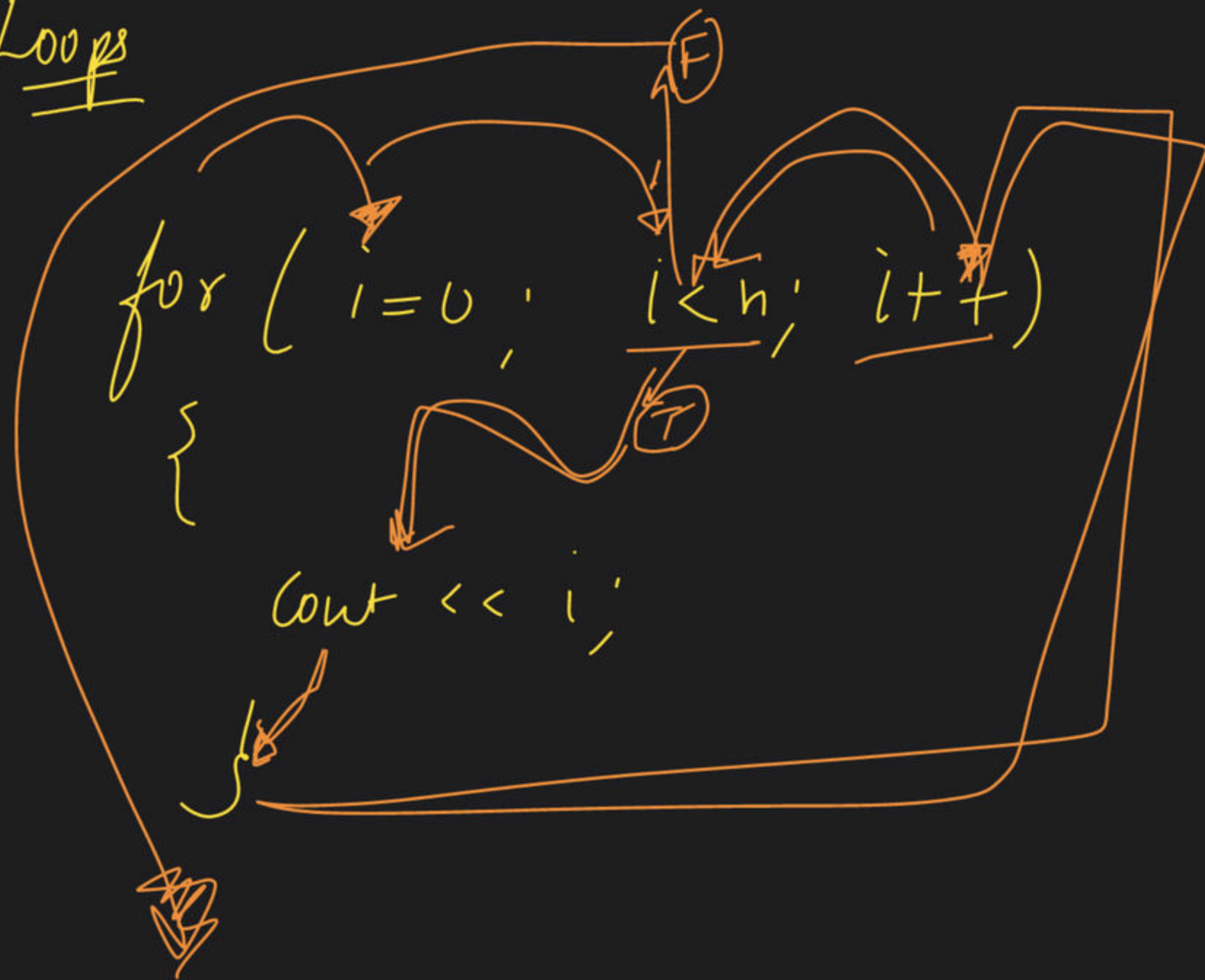




L1-Pattern Printing

Special class

① Loops



② Nested Loops

outer loop

~~inner loop~~

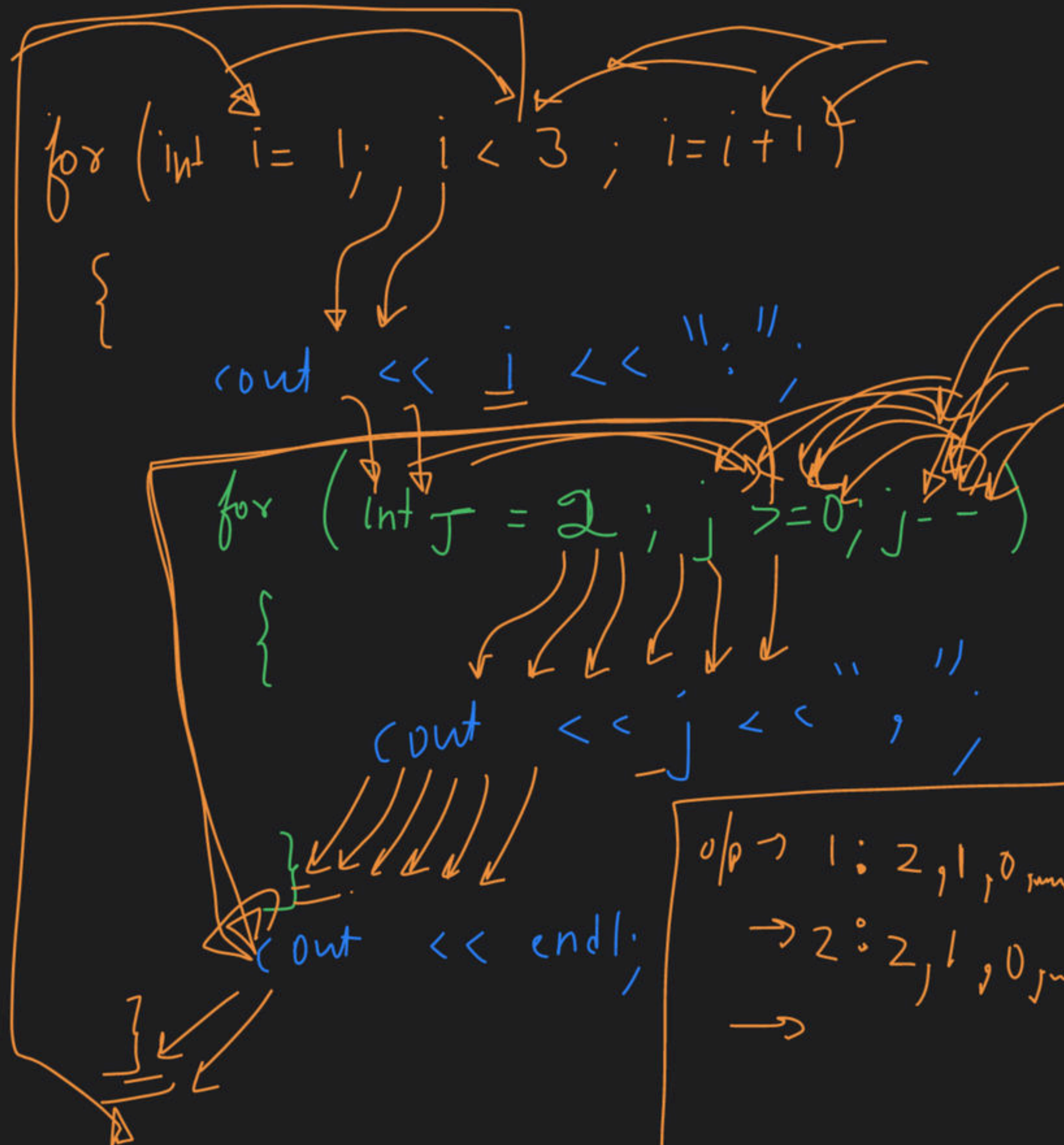
o/r →

0 : 1, 2, 3

$$\begin{aligned} \rightarrow 1 &: 1, 2, \dots \\ \rightarrow 2 &: 1, 2, \dots \end{aligned}$$
$$i = 0$$
$$b \leq 3 \rightarrow +$$
$$1 < 3 \rightarrow +$$
$$\bar{j} = 2$$
$$2 < 3 \rightarrow T$$
$$J=3$$
$$3 < 3 \rightarrow F$$
$$1 \leq 3 \rightarrow T$$



outer
{ single / or / -
inner {
}
}



i = 1
1 < 3 → T

j = 2

2 >= 0 → T

j = 1

1 >= 0 → T

j = 0

0 >= 0 → T

j = -1

-1 >= 0 → F

i = 2
2 < 3 → T

j = 2

2 >= 0 → T

j = 1

1 >= 0 → T

j = 0

0 >= 0 → T

j = -1

-1 >= 0 → F

i = 3
3 < 3 → F




```
for (int j = 3; j > 0; j = j - 1)
{
```

```
    for (int i = 0; i < 10; i = i + 3)
    {
```

```
        cout << (i + j) << " , "
```

```
    }
    cout << endl;
}
```

o/p:

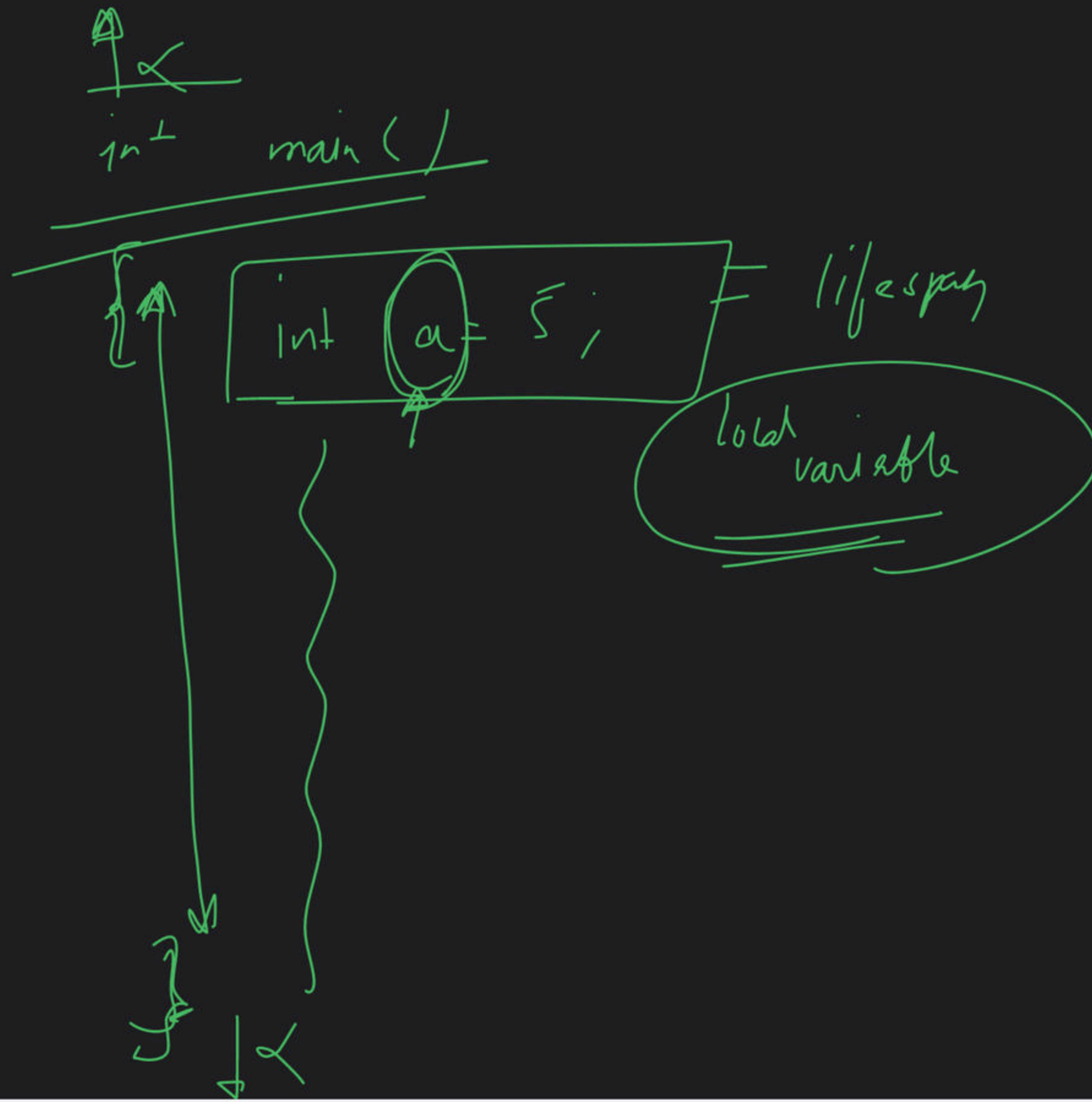
```
→ 3, 6, 9, 12
→ 2, 5, 8, 11
→ 1, 4, 7, 10
```

j=3	j=2	j=1	j=0
3 > 0 → T			
i=0	i=0	i=0	0 > 0-1 → F
0 < 10 → T			
i=3	i=3	i=3	
3 < 10 → T			
i=6	<u>i=6</u>	i=6	
6 < 10 → T			
i=9	i=9	i=9	
9 < 10 → T			
12	<u>i=12</u>	i=12	
i=12	↓	↓	
12 < 10 → F	F	F	

Variable

~~for~~ scope

30
↳ not a loop



Local
variable

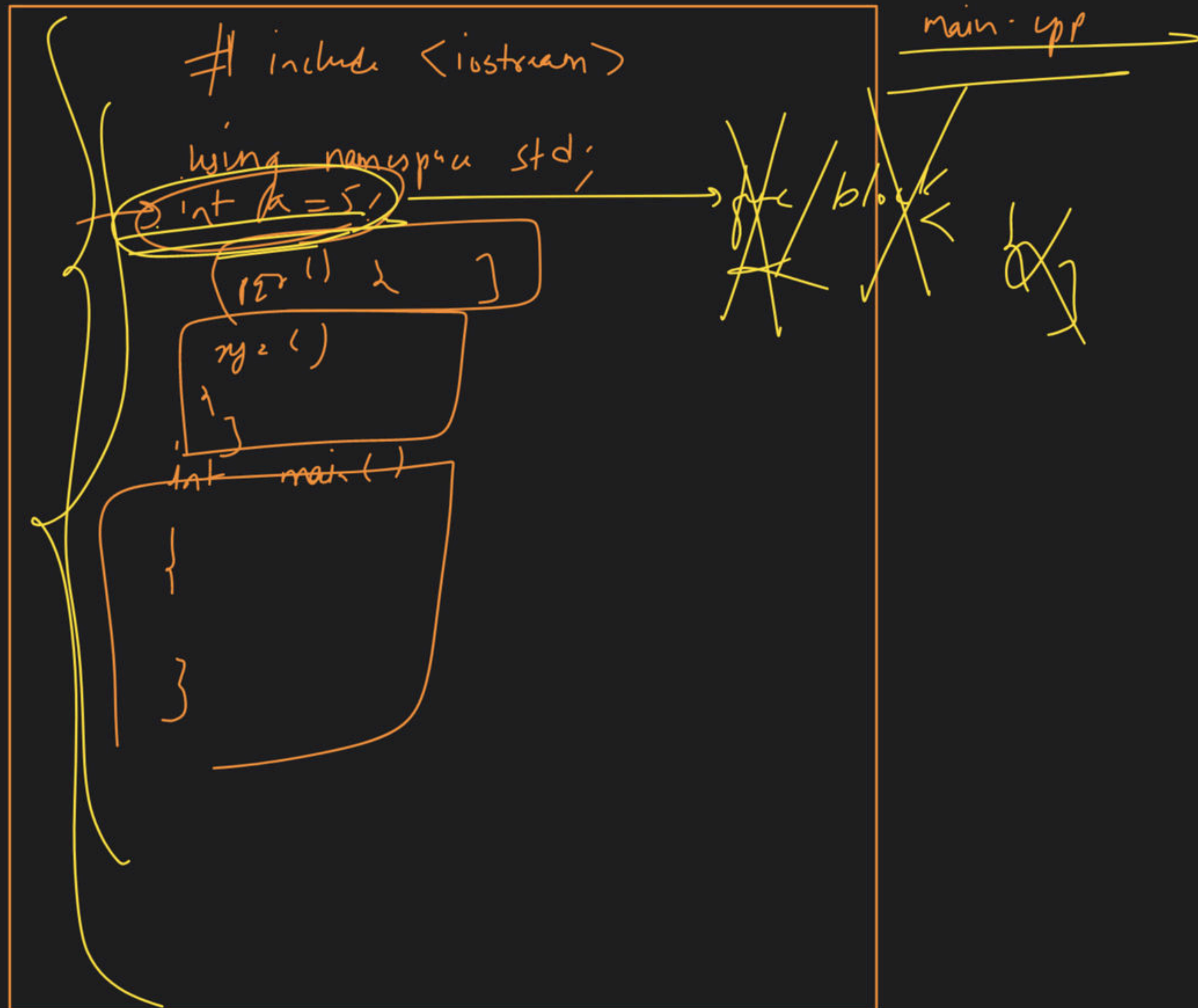
myz()

{
int a = 5;

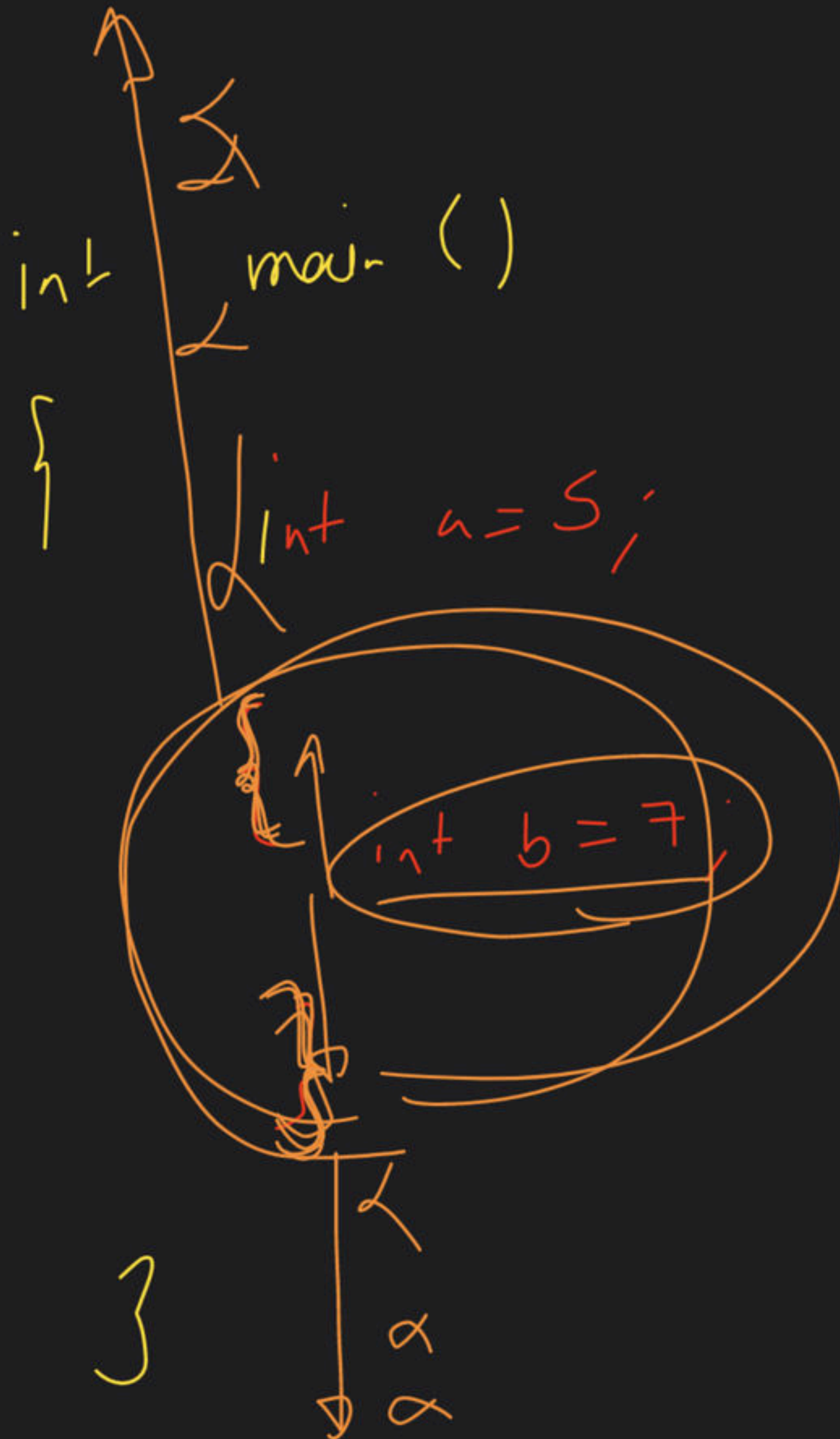
}

Global
variable

Why → ?



Block-scope
Variable



```

int a = 5;
cout << a;
int a = 6;
cout << a;

```

Lyro


```
int main()
```

```
{
```

```
int a = 5;
```

```
{ int a = 6;
```

```
}
```

```
}
```

370



a

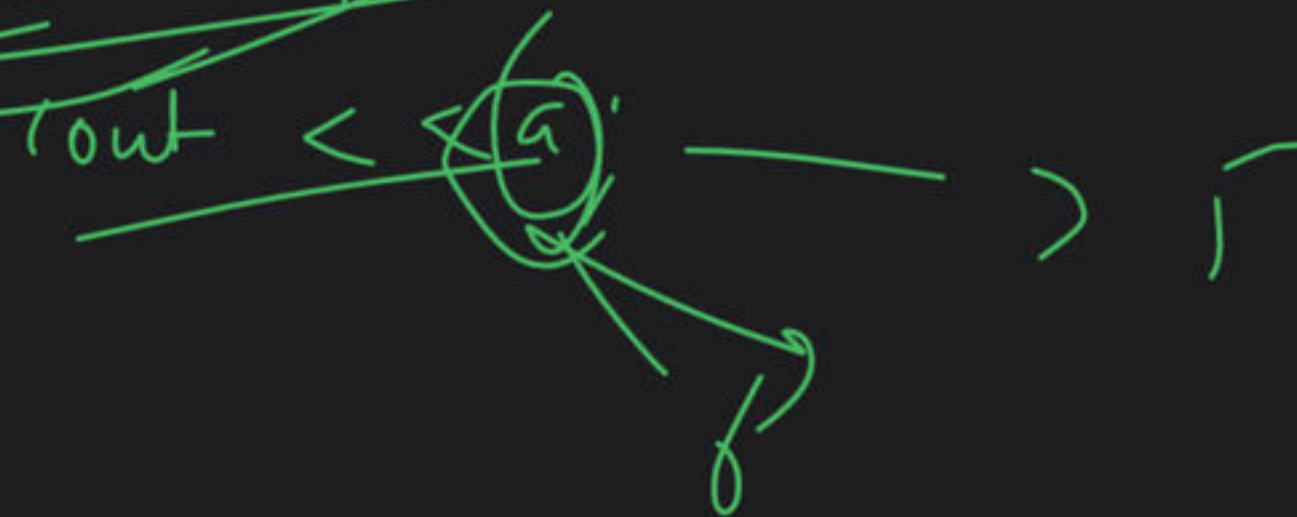


a

718

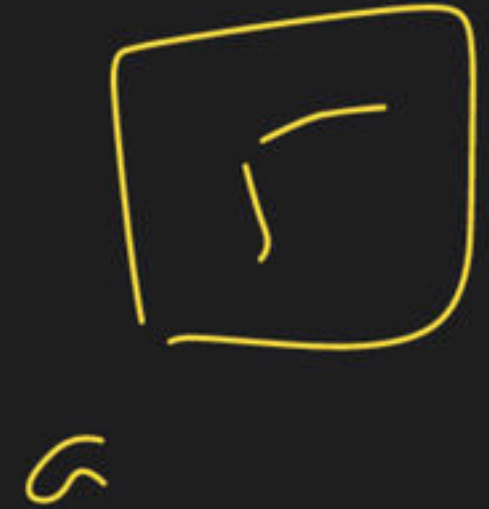


c



```
int main()
```

```
{  
    int a = 5;  
}
```



~~int a = 6;~~ → Error

← } cout << a;

int bubble()

```
{  
    int a;  
    cout << a;  
}
```

int main()

```
{  
    int a = 5;  
}
```

H/w

sizeof()

D+

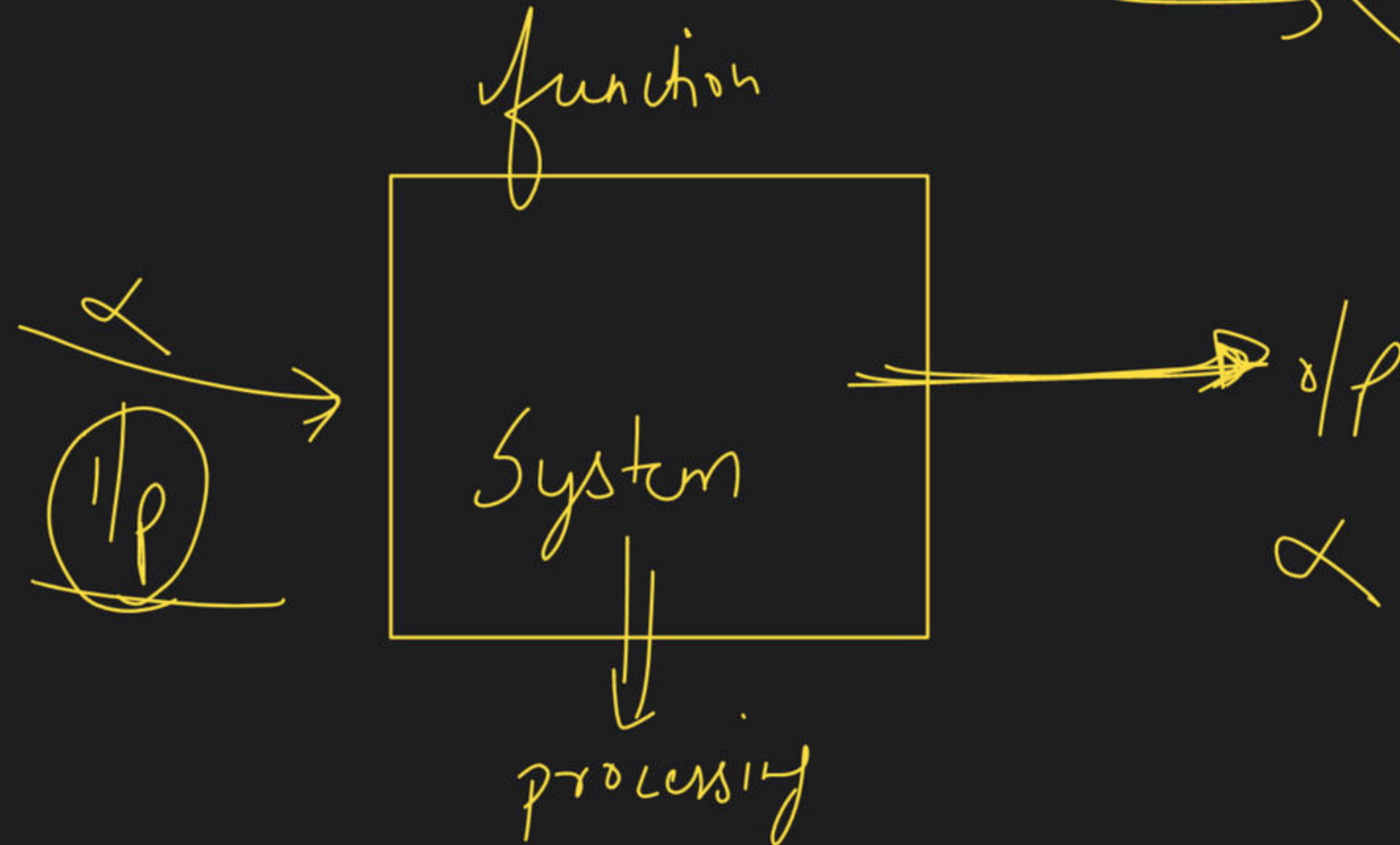
↳ int

↳ 4 byte

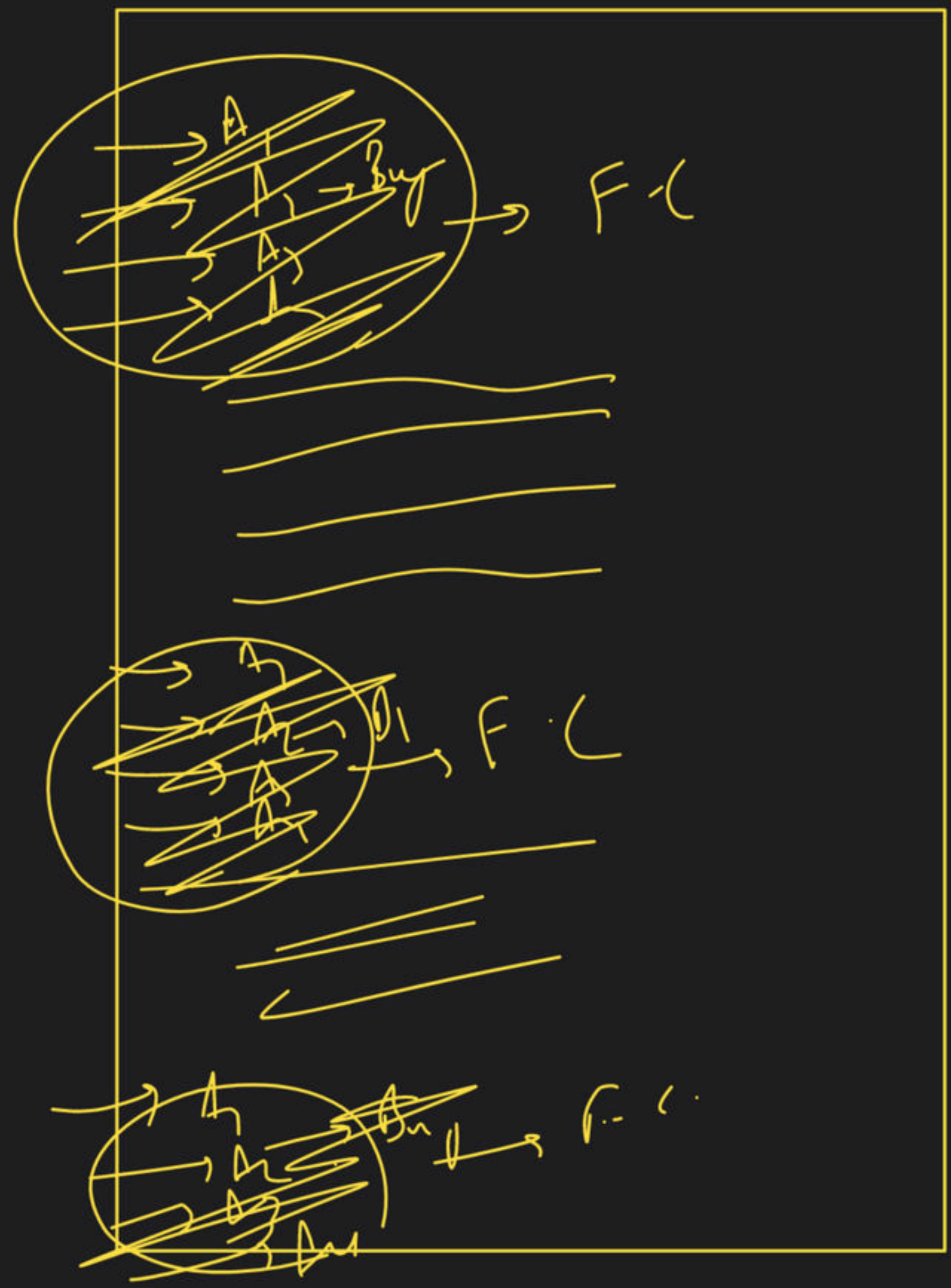
int a = 5;

cout << sizeof(a);

function :-



output
out
schuen



func)
{
 A
 A
 A
}

Function



Syntax →

return type

function_name

(int a, int b)

parameter

{

~~~~~

~~~~~

~~~~~

} processing

return value

}



function

print Babbar / Sum of Number

(convert Into Upper Case)

Void → Kuch return nhi Krnaga

declare

Void printBabbar ();

int sum of Number (int a, int b);

char convertIntoUpperCase (char c);

Why we need it?

H/w

function define

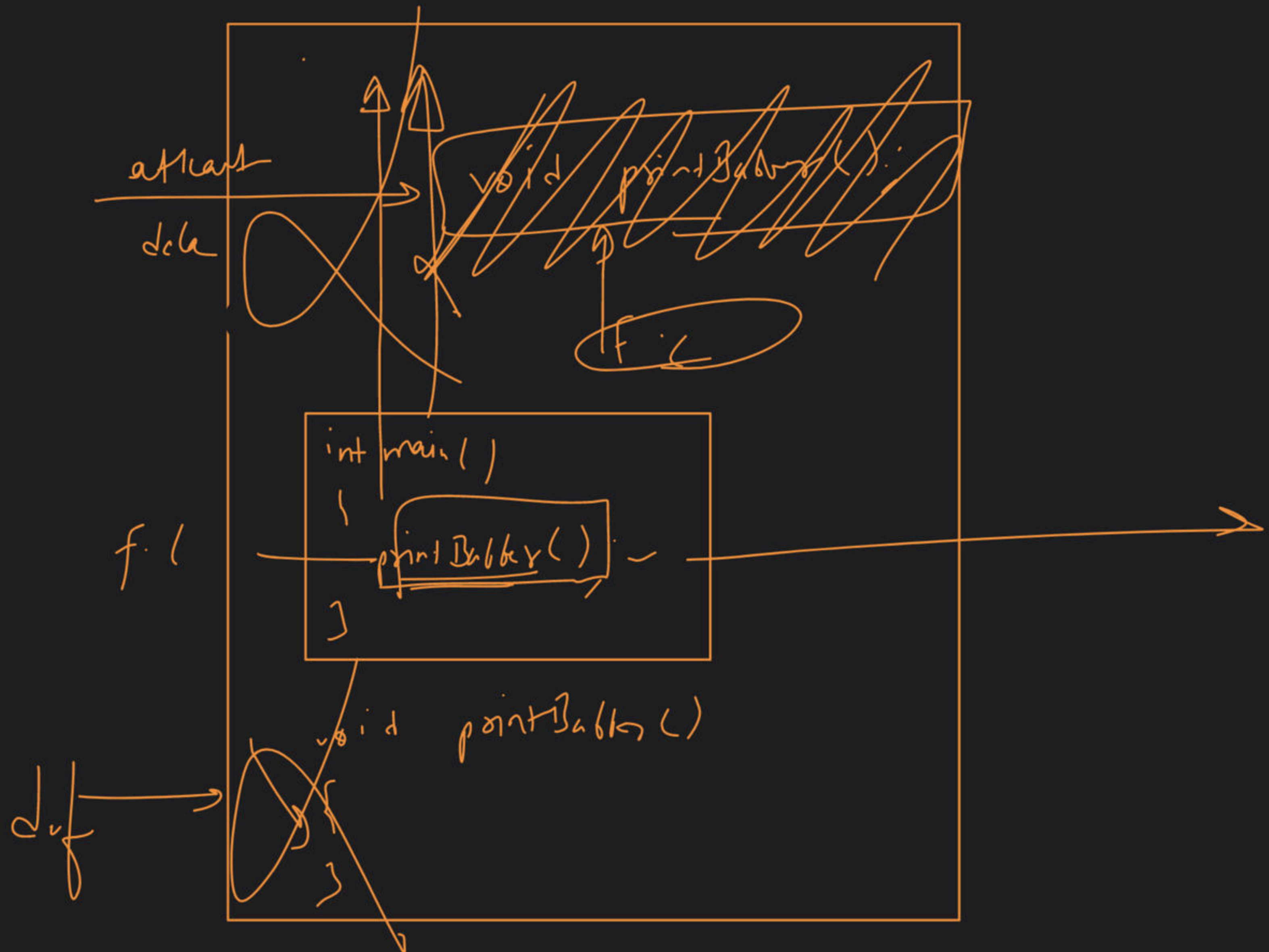
use → function call

```
void printBubble()  
{  
    cout << "Bubble";  
}
```

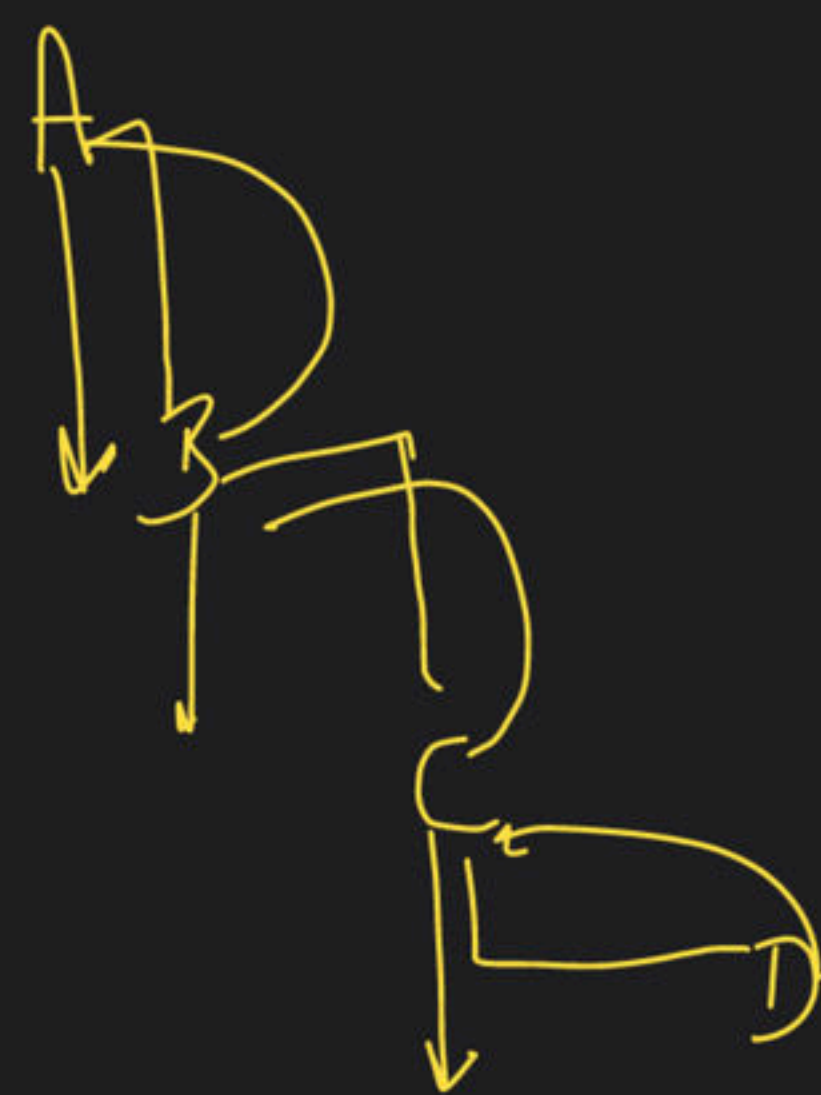
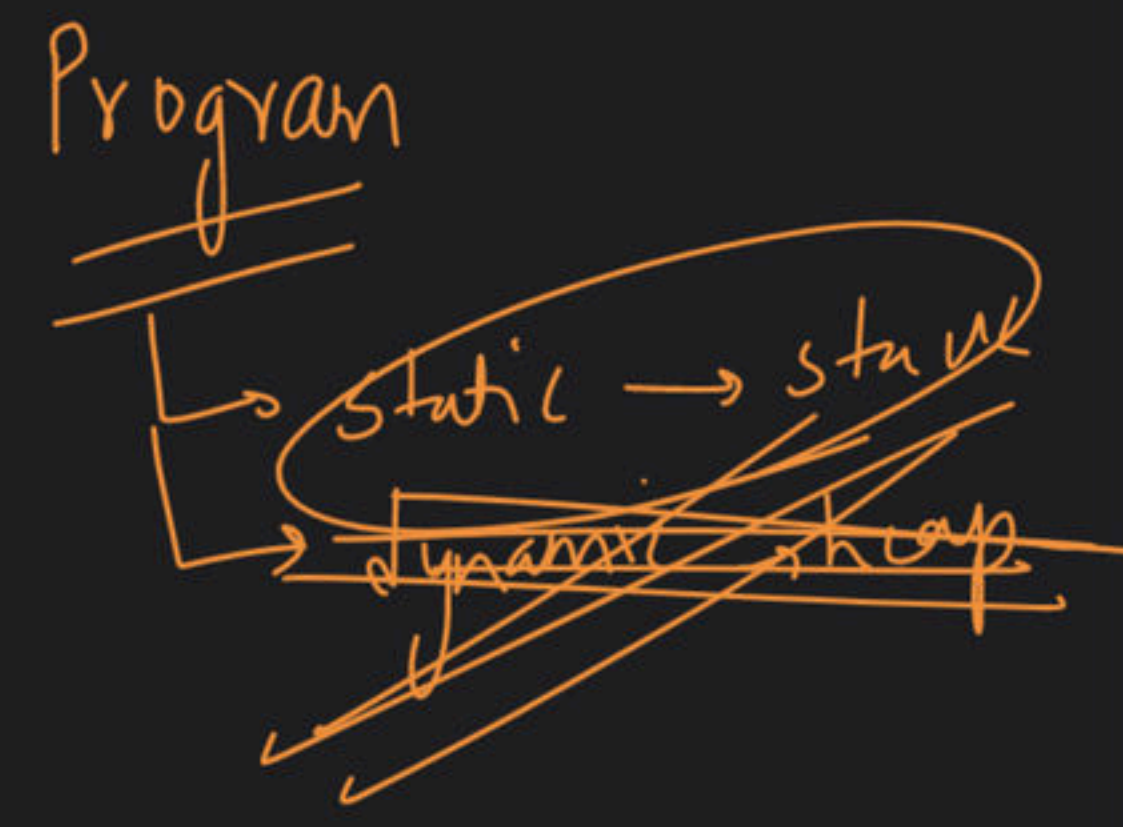
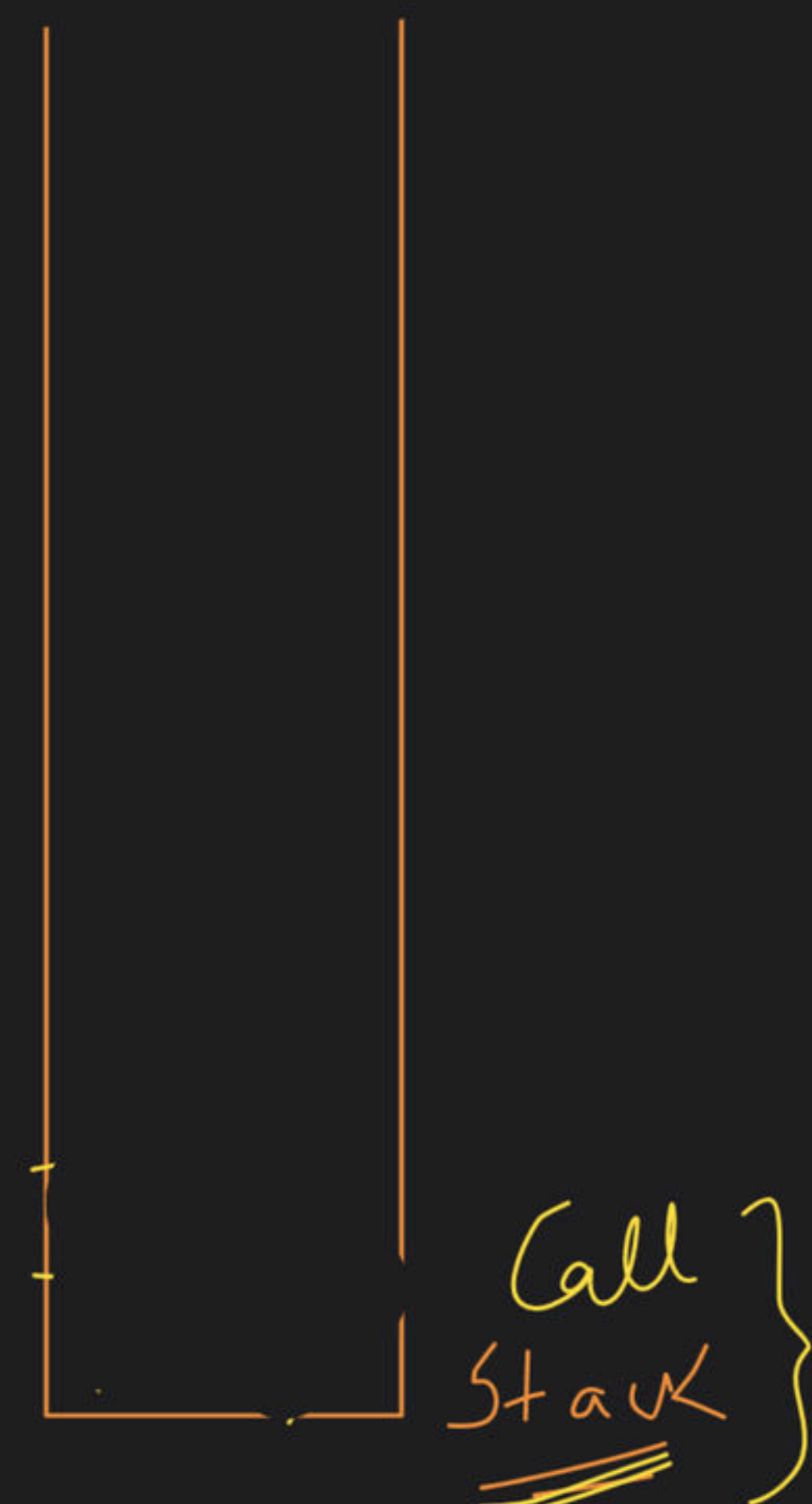
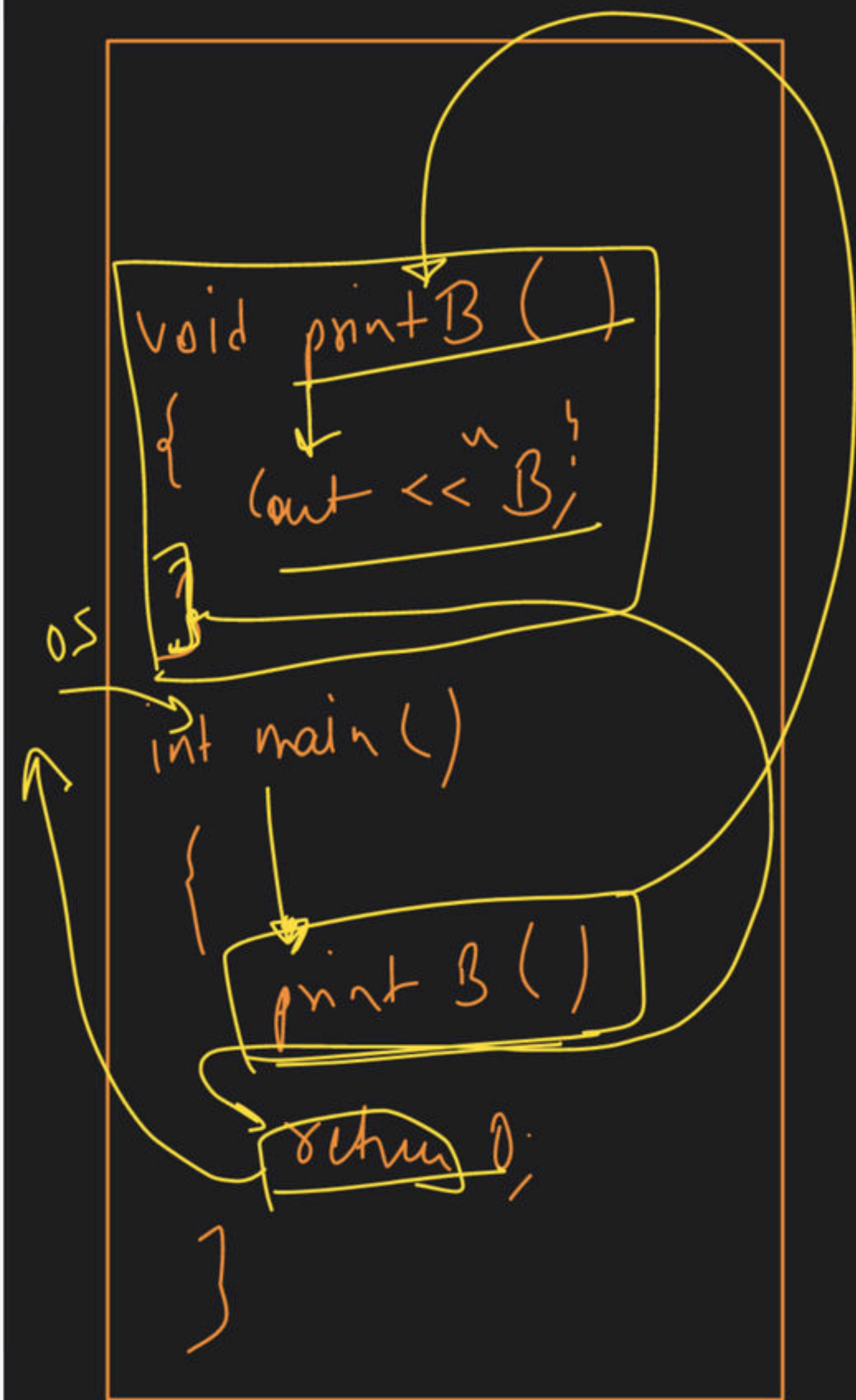
argument  
↓  
printBubble(3,4)

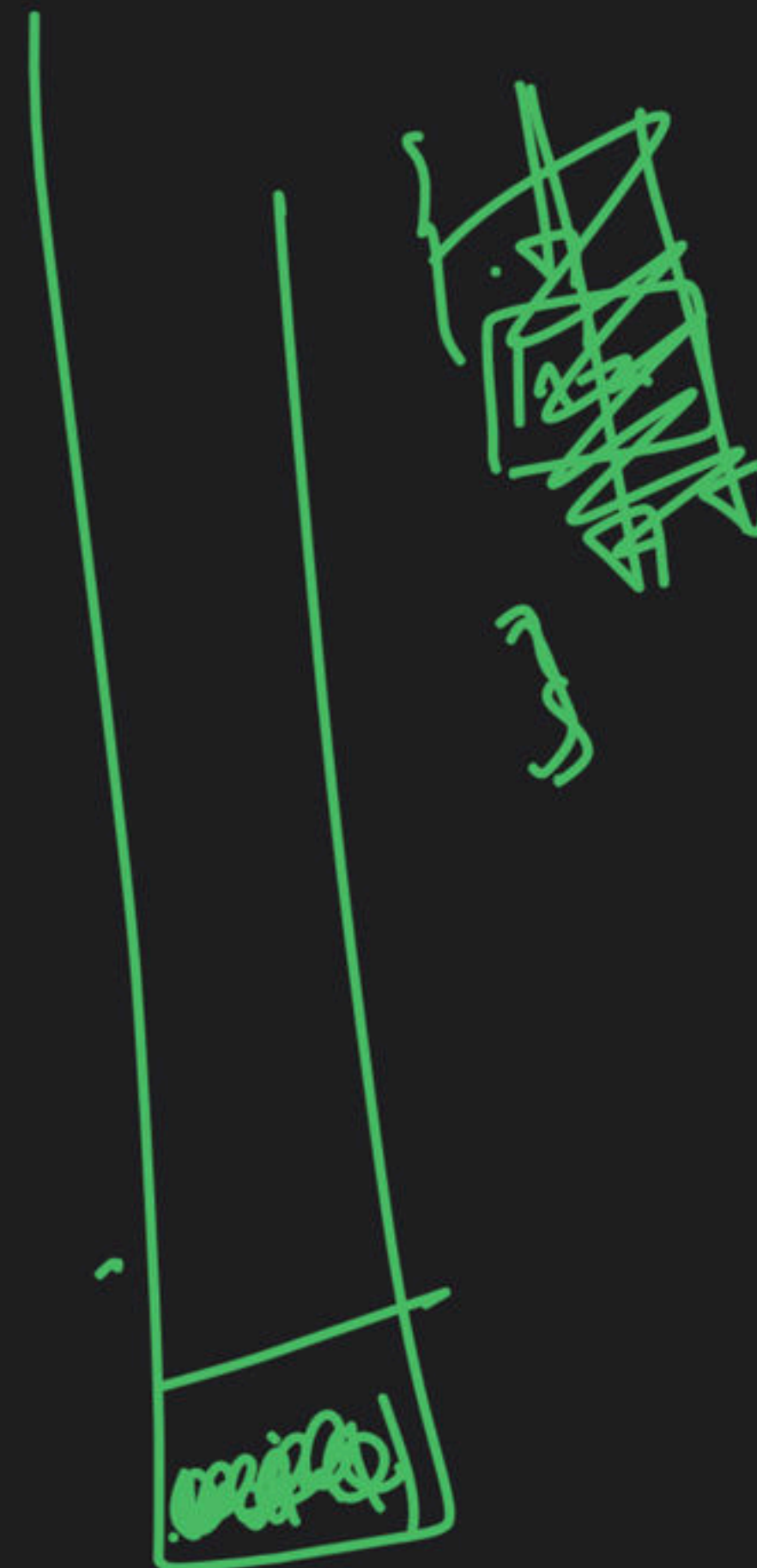
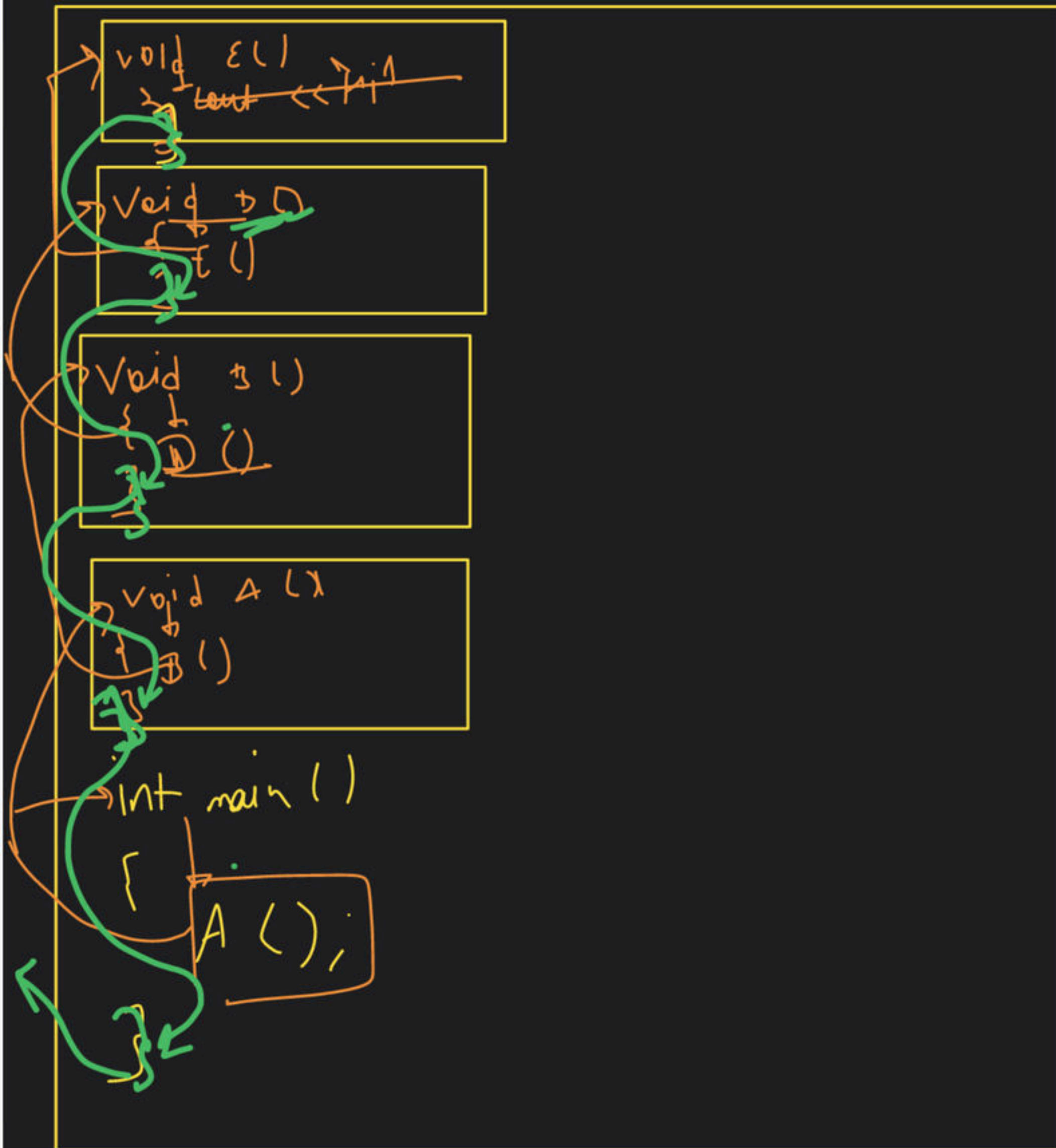
FL → 316 → printBubble();













① ~~if~~ num  $\rightarrow$  { even  
odd

② true / -ve

①  $1 \rightarrow N$  count

⑥ 2 km + ash

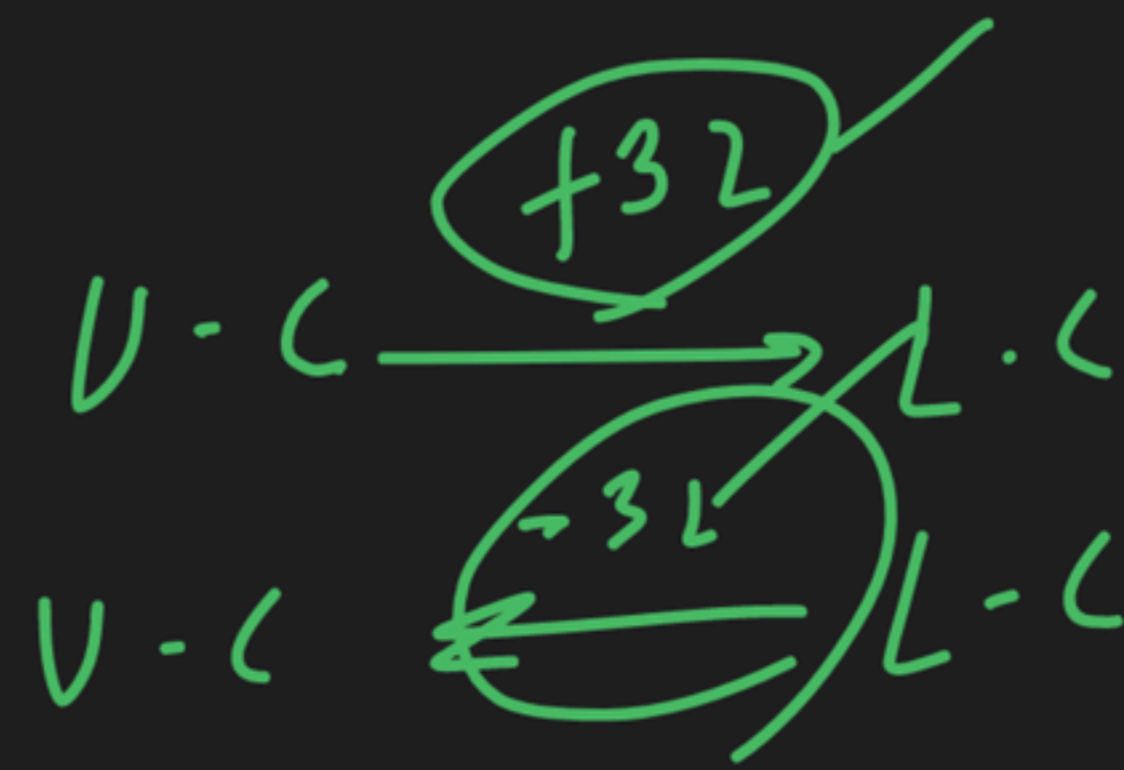
0 Average

convert To VC

$A \rightarrow 65$   
 $a \rightarrow 97$

convert To LC

37  
31 32



$65 \xrightarrow{+32} 97$

$V.C \longrightarrow L.C$   
 $\begin{matrix} 65 \\ A \end{matrix} \longrightarrow \begin{matrix} 97 \\ a \end{matrix}$



$$\underline{A} \longrightarrow \underline{a}$$

$$\cancel{A} - \cancel{A} + a \rightarrow (a) \quad D - \cancel{A'} + \cancel{a'} \rightarrow ?$$

$$U.C \xrightarrow{-A + a'} L.C$$

$$A \rightarrow Z$$

$$a \rightarrow z$$

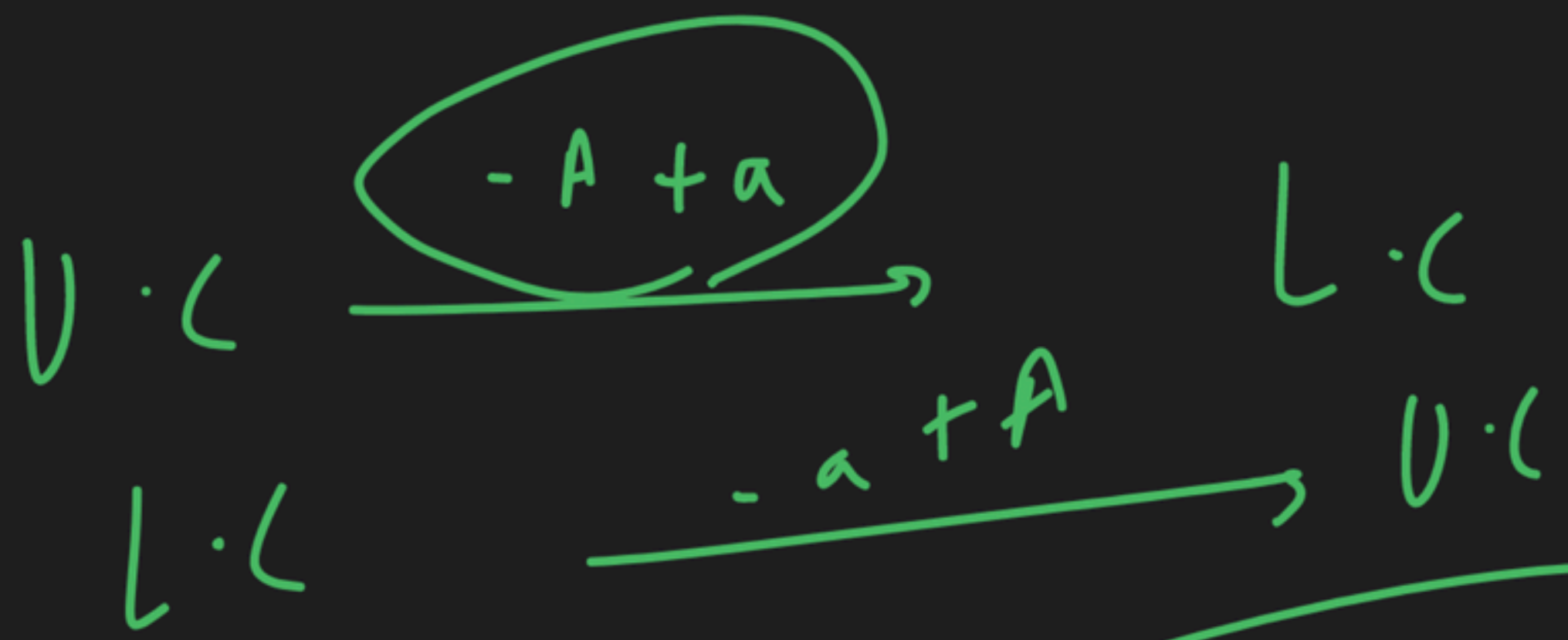
$$\cancel{A} - \cancel{A} + a \rightarrow a$$

$$B - A + a \rightarrow ?$$

$$C \rightarrow A + a \rightarrow ?$$

$$3 \longrightarrow 10$$

$$\cancel{3} - \cancel{3} + 10 \longrightarrow 10$$



$66 - 65 + 97$   
 $1 + 97$   
 $98$

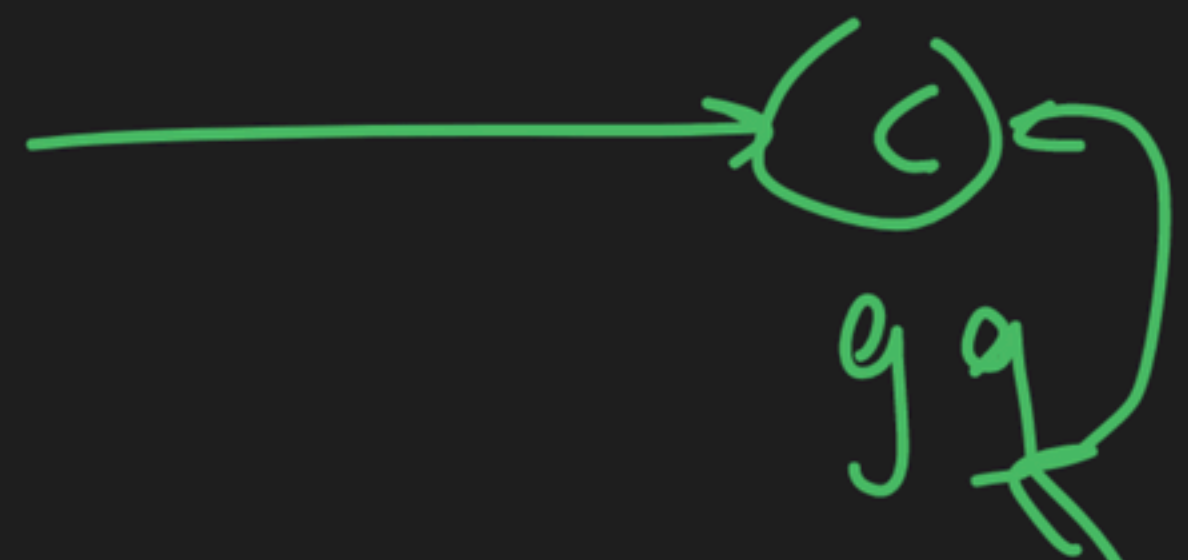
$B \rightarrow 66$

$\rightarrow b \leftarrow$   
 $\downarrow$   
 $98$

$-A+a \rightarrow 32$   
 $-65+97 \rightarrow 32$



C  
67



C  $-A' + a'$

67 - 65 + 97

2 + 17



Tum  
galt  
Kampf

~~A - A' - a'~~  
~~B - B' - b'~~  
~~C - C' - c'~~

$n \rightarrow y$   
 $(n - n + y)$

$\{$   $\rightarrow$   $e$   
 $(0)$   
 $69 - A + a$   
 $69 - 15 + 97$

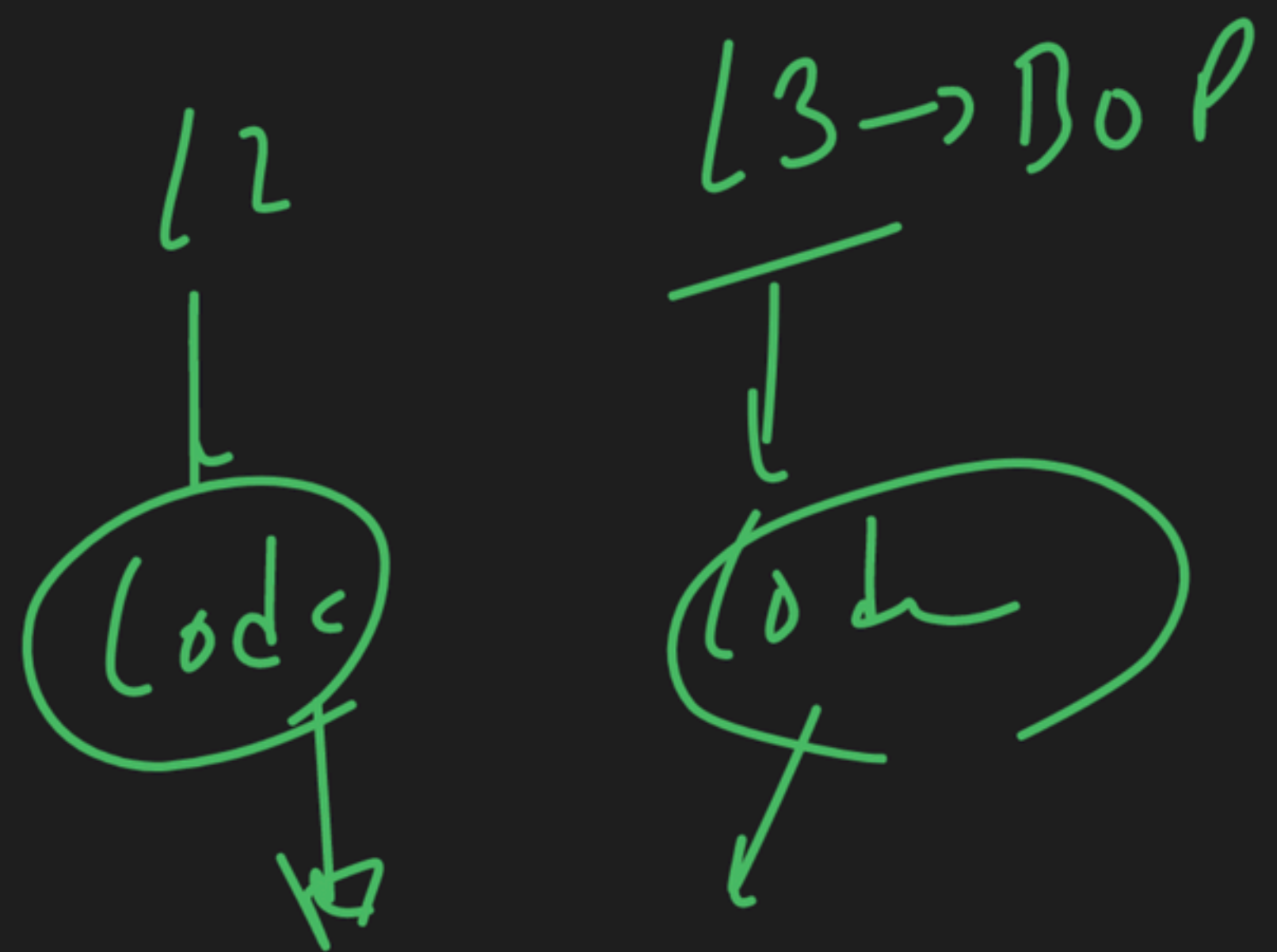
Math

$5 \rightarrow 15$   
 $(5 - 5 + 15)$

32

$(A + a)$

|     |     |       |                  |
|-----|-----|-------|------------------|
|     | 65  | 42    |                  |
| $=$ | $A$ | $+ a$ | $\rightarrow 32$ |
| $-$ | $B$ | $+ b$ | $\rightarrow 32$ |
| $-$ | $C$ | $+ c$ | $\rightarrow 32$ |
| $-$ | $D$ | $+ d$ | $\rightarrow 32$ |

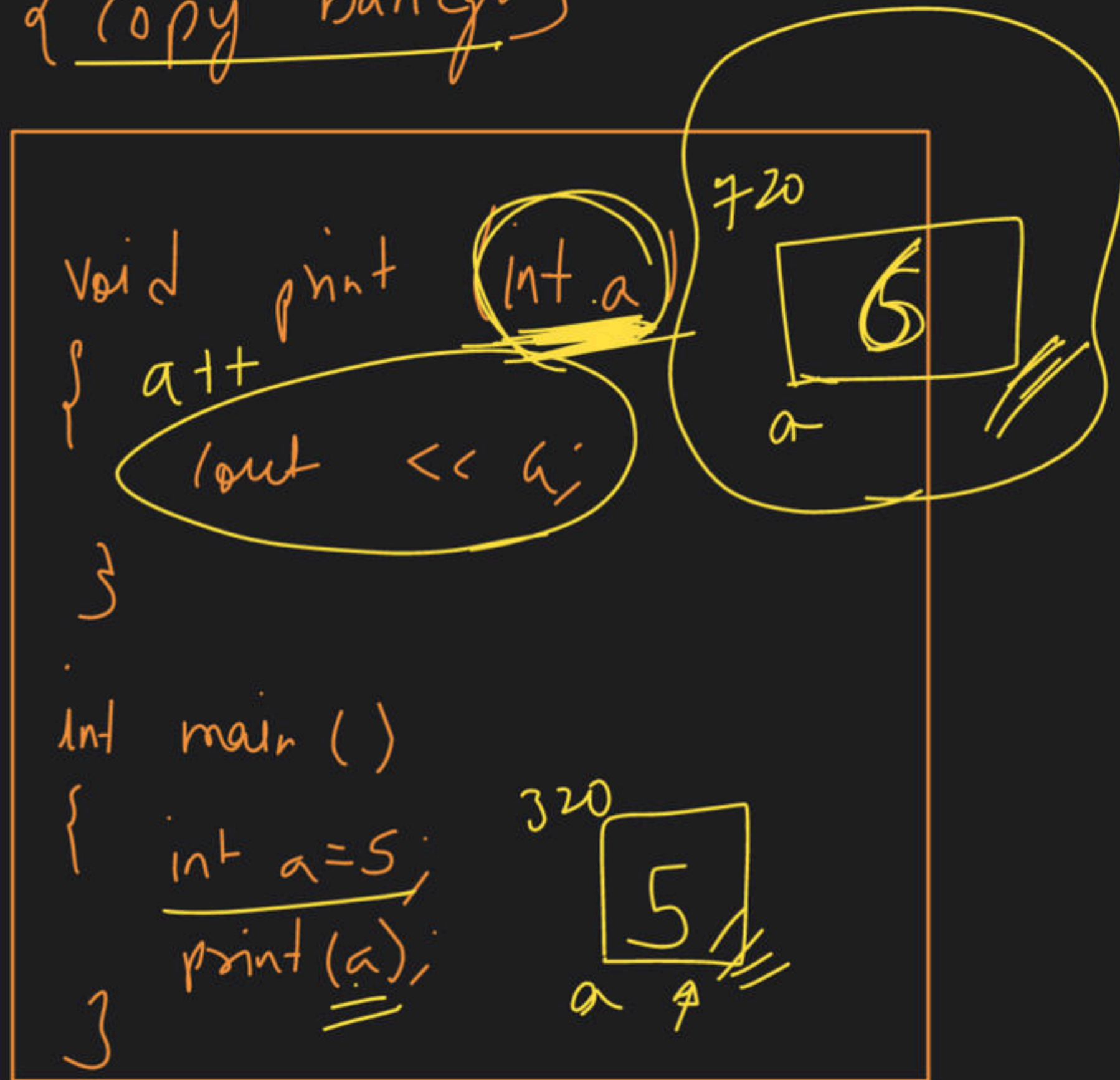


touchabbar 3@gmail.com



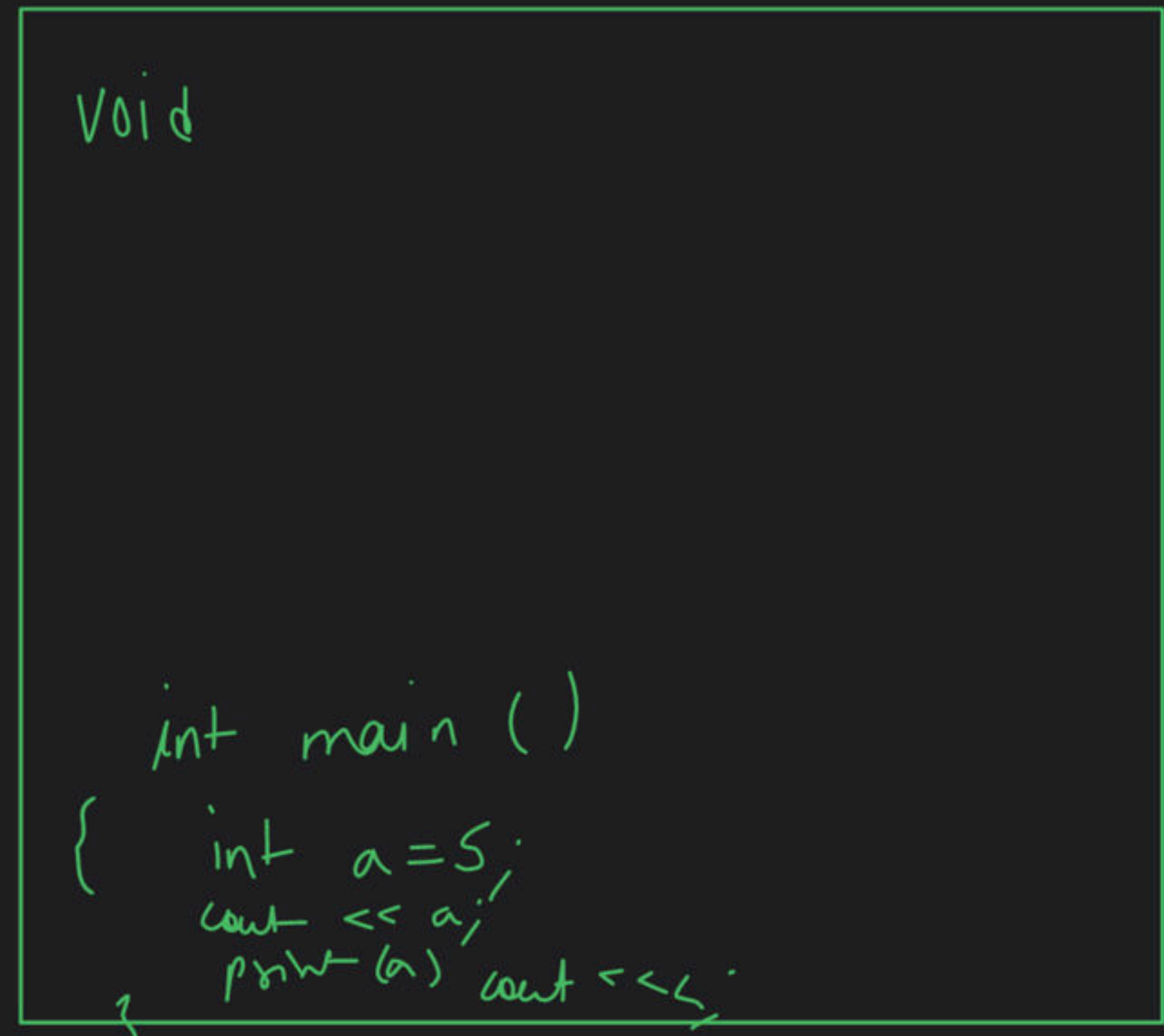
# Pass by value

{ copy banegi }



# Pass by reference

{ actual value me change }



```
void print (int a)
```

720



```
{ cout << a;
```

```
  a++;
```

```
  a++;
```

```
  a++;
```

```
  cout << a;
```

```
  a--;
```

```
}
```

```
int main()
```

```
{ int a=5;
```

```
  a++;
```

```
  cout << a;
```

```
  print(a);
```

```
  cout << a;
```

320



o/p →

6  
6  
9  
8



void print

(int &a)  
int &b



~~cout << a;~~

a++;

b = a \* 2

b = a \* 2

b++

return;

int main()

{ int a = 5;

a++;

cout << a;

print(a)

cout << a;

320



o/p

int &a

6

6

29





```
void print (int &love)
```

```
{
```

```
    love = love * 10
```

```
    love ++
```

```
    love ++
```

```
}
```

```
}
```

```
int main ()
```

```
{    int a = 5
```

```
    cout << a;
```

```
    print(a);
```

```
    cout << a;
```

love

52

|    |
|----|
| 52 |
| 52 |
| 52 |

a

u/p

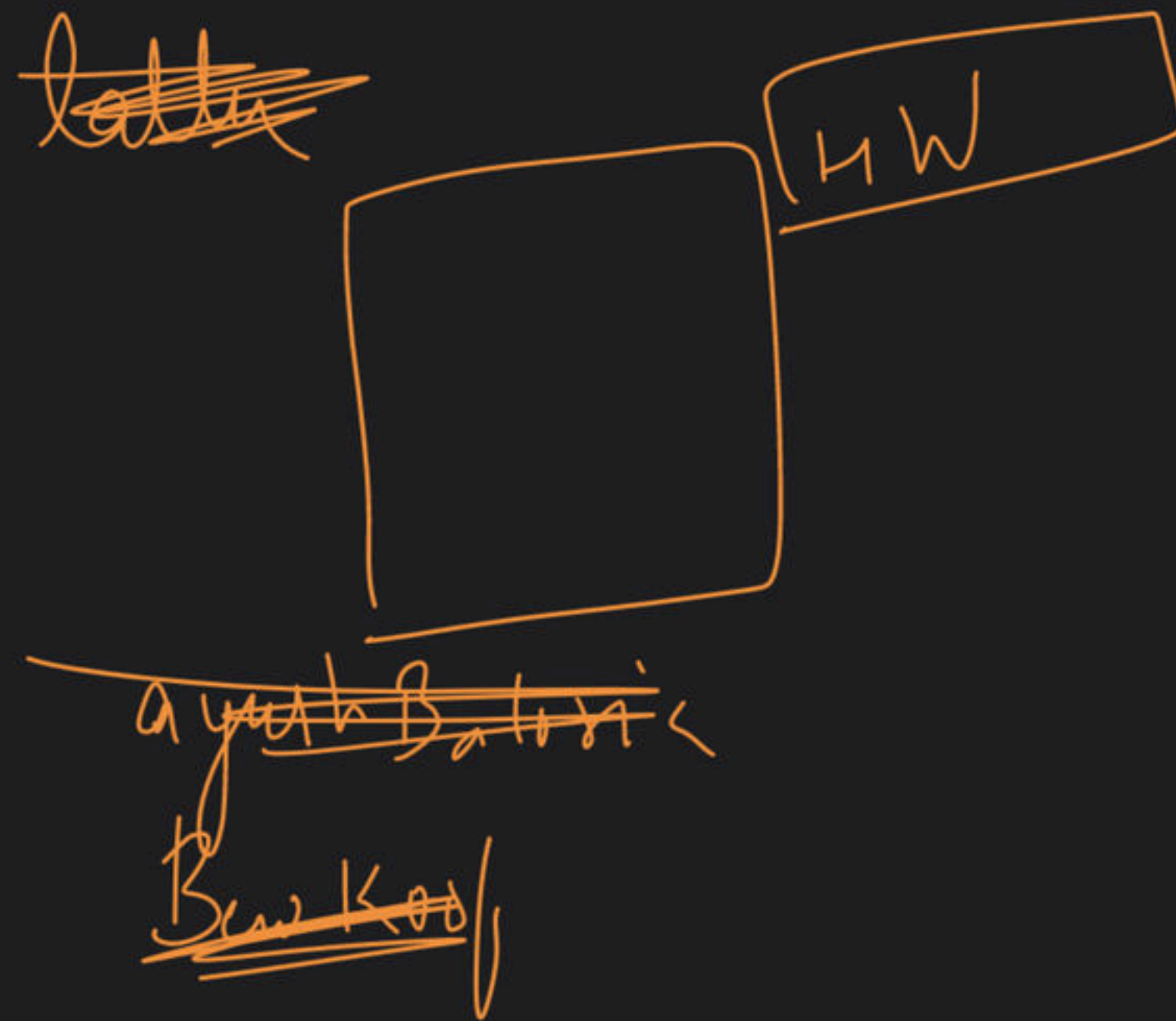
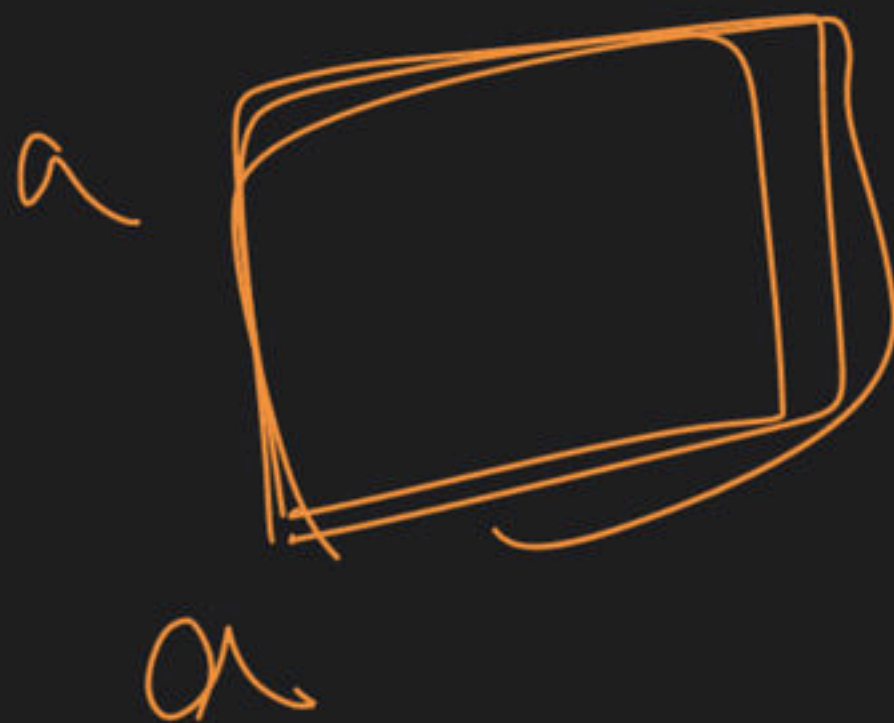
→

5

52

int &c

int &c







```
void modify (int &a)
```

```
{  
  a = a + 10;  
}
```

```
}
```

```
int main ()
```

```
{
```

```
  int n = 5;
```

```
  modify (n);
```

```
  cout << n;
```

```
}
```

a  
15  
n

15

```
void print (int a, int b)
```

```
{  
    a = a * 10;  
    b = b + 10;  
}
```

50  
~~5~~  
a

```
int main ()
```

```
{  
    int a = 5;  
    int b = 10;  
    print(a, b)  
    cout << a << " " << b  
}
```

5  
a

20  
b

5 20

5 2  
5 10  
5 15  
5 20



Sonayna

Wednesday



En

10 different

A



Nested  
Loop

Patton Printing



1

